5.6 Food chains

Science understanding

Verbal/Linguistic Visual/Spatial

A food chain shows the feeding relationships in an ecosystem. In the following food chain, letters are used to represent the organisms.

A → B → C → D

1 **Explain** what the arrow in a food chain represents.

_____

2 **Identify** the letter representing a:

(a) producer organism _____

(b) herbivore _____

(c) first order consumer _____

(d) carnivore _____

3 **Construct** food chains from the following information.

(a) The hawk circled overhead before swooping down on the snake. Earlier that day, the snake had eaten a desert rat that was eating grass.

swoop (v) to quickly and suddenly fly downwards, possibly to attack



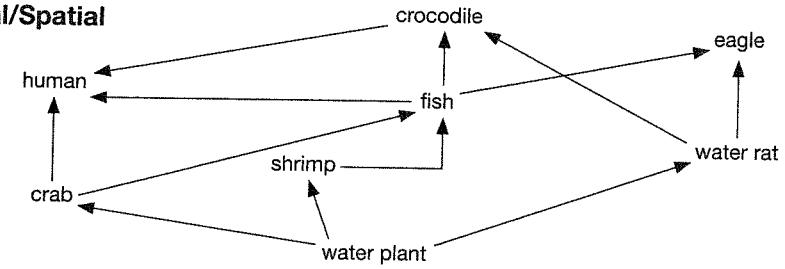
(b) A fat green caterpillar was eating a leaf. A butcher bird was on the branch above, about to eat the caterpillar.

(c) Killer whales swim through the ocean looking for seals to eat. In turn, seals search for large fish that have grown fat on their meals of smaller fish. Floating in the water are millions of plankton. These are microscopic producer organisms that provide food for small fish and other herbivorous (plant-eating) animals in the ocean.

5.7 Food webs

Science understanding

Visual/Spatial



Use the food web above to complete the following tasks.

1 **Construct** two food chains from the food web.

2 **Identify** the herbivores in the food web.

_____

3 **Identify** the organisms that are in competition for the water rat as a food source.

_____

4 The human is a second order consumer when it eats the crab. **Deduce** what order of consumer the human is when it eats:

(a) the fish _____

(b) crocodile meat _____

5 **(a)** **Propose** what would happen to the fish population if the water rat disappeared from the area.

_____

(b) **Justify** your proposal.

_____

6 **(a)** Based on your answers to question 5, **propose** what would then happen to the crab and shrimp populations.

_____

(b) **Justify** your proposal.

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